COMP 2121 – Quiz 2 Time: 20 min. Possible marks: 5	Name: Student #:

1. Use rules of inference to prove that the following argument is valid.
   Note: You must use the provided rules of inference and list the rule. No marks will be given for proofs that involve a truth table. *(2.5 marks)*

$x \wedge y$
$x \rightarrow \neg(y \wedge z)$
$w \rightarrow z$

$\therefore \neg w$

	STEPS	REASON
1)	$x \wedge y$	given
2)	x	C.S. 1
3)	y	C.S. 1
4)	$x \rightarrow \neg(y \wedge z)$	given
5)	$\neg(y \wedge z)$	M.P. 2,4
6)	$\neg y \vee \neg z$	De Morgan's 5
7)	$\neg z$	D.S. 3,6
8)	$w \rightarrow z$	given
9)	$\neg w$	M.T. 7,8
10)		
11)		
12)		
13)		

2. Prove if each of the following statements are TRUE or FALSE. Note that you must provide justification for your answers using a table of values. No marks will be given for answers that use other methods of justification or that provide no justification.

The universe for x is {-2, 0, 6}.
The universe for y is {-2, -1, 0, 1}

(a) *(0.5 marks)* $\forall x, \forall y,\ x > |2y|$ TRUE / FALSE (circle)

x	y	$x > \|2y\|$
-2	0	$-2 > 0$ ✗

(b) *(1 mark)* $\forall x, \exists y,\ x > |2y|$ TRUE / FALSE (circle)

x	y	$x > \|2y\|$
0	-2	$0 > 4$ ✗
0	-1	$0 > 2$ ✗
0	0	$0 > 0$ ✗
0	1	$0 > 2$ ✗

(c) *(1 mark)* $\exists x, \forall y,\ x > |2y|$ TRUE / FALSE (circle)

x	y	$x > \|2y\|$
6	-2	$6 > 4$ ✓
6	-1	$6 > 2$ ✓
6	0	$6 > 0$ ✓
6	1	$6 > 2$ ✓